Question and Dataset

1. Organization

Australian Competition and Consumer Commission (ACCC) — Australia's national competition and markets regulator. The ACCC publishes regular monitoring reports on retail fuel competition and has flagged ongoing concerns about market concentration and pricing dynamics in Australian fuel retailing. It has a standing interest in whether new entrants — particularly hypermarket-style discount entrants like Costco — meaningfully discipline incumbent pricing, since lowering entry barriers is one of the primary tools available to the ACCC for improving competition in concentrated markets.

2. Research question / classification

Does the entry of a Costco gas station into an Australian local market cause nearby competitor stations to lower their retail fuel prices?

This is a **causal** question. We will answer it using **synthetic control**, applied separately to each of four Costco openings with adequate pre- and post-treatment data. The synthetic control method — one of the four identification strategies described in the project specification — is explicitly recommended for "a single treated unit or a handful of treated units, a clear treatment date, and a large donor pool of comparable untreated units with pre-treatment data," which describes our setup precisely. Difference-in-differences was considered first but its parallel-trends identifying assumption fails empirically in our data; synthetic control does not require parallel trends and is the methodologically appropriate choice.

3. Dataset

We combine three publicly accessible Australian state fuel-price registries into a unified station-month panel: **NSW FuelCheck** (maintained by the NSW Government, mandatory station reporting since August 2016, accessed via api.nsw.gov.au and data.nsw.gov.au), **Queensland Fuel Price Reporting Scheme** (QLD Treasury, mandatory reporting since December 2018, accessed via data.qld.gov.au), and **WA FuelWatch** (WA Government, daily mandatory reporting since January 2001, accessed via data.wa.gov.au).

We pull historical snapshots from each registry covering January 2018 through April 2026, yielding approximately **4.4 million station-day price records** across **~6,000 unique stations** in the three states. Each row in the raw data is a station × day × fuel-type price observation. We aggregate to monthly mean prices per station for the analysis. To this panel we merge a hand-collected list of Costco gas station opening dates, verified against each station's first-appearance date in the relevant registry. Key fields are listed below.

Field	Description
`state`	State (NSW, QLD, or WA)
`station_id`	Unique station identifier (state-scoped)
`name`	Service station name

`brand`	Brand affiliation (Costco, Caltex, Shell, BP, independents, etc.)
`address`	Street address
`latitude / longitude`	Geographic coordinates (used to compute haversine distance to Costco)
`postcode / suburb`	Postcode and suburb — used to define donor pool units
`date / year_month`	Date of price observation; aggregated to month for the panel
`fuel_type`	Regular unleaded (ULP / U91 / E10), premium, or diesel
`price`	Retail price at the pump (cents per liter)

4. Key components

Method. Synthetic control, applied separately to each of four Costco openings. For each treated Costco we construct a synthetic counterfactual as a weighted combination of donor postcodes whose pre-treatment price trajectory matches the treated market's pre-treatment trajectory. The post-treatment gap between the actual treated market and the synthetic counterfactual is the estimated treatment effect.

Treatment. The opening of a Costco gas station within a local market. Treatment switches on at the Costco station's first-observed date in the relevant state's price registry and remains on thereafter.

Treated units. Four Costco markets, each defined as the monthly mean unleaded price across all non-Costco stations within 5 km of that Costco's coordinates:

Costco	**State**	**Treatment date**	**Pre months**	**Post months**
Coomera	QLD	May 2023	51	31
Casuarina	WA	November 2022	58	42
Perth Airport	WA	April 2020	27	73
Lake Macquarie	NSW	May 2022	58	35

Australia has six additional Costco fuel stations (Marsden Park, Auburn, Canberra Airport, Casula, North Lakes, and Ipswich), but these have insufficient pre- or post-treatment data for the method — either the warehouse opened before its state's registry began reporting (Casula, North Lakes, Ipswich), the postcode lacked any pre-period coverage (Canberra Airport), the post-treatment window is too short (Auburn, just opened August 2025), or the pre-period is too short for stable weight fitting (Marsden Park). These stations appear in the project as descriptive context but are excluded from the formal estimation.

Donor pool. Each Australian postcode in the three states with at least 3 average unique stations per month, at least 24 months of observed coverage, and located more than 20 km from *any* Costco fuel station (whether treated or excluded) becomes one donor unit, defined as the postcode's monthly mean unleaded price across all stations in that postcode. After applying these filters the donor pool contains **196 postcodes**: 104 in NSW, 56 in QLD, and 36 in WA, yielding 17,578 (postcode × month) panel observations. For each Costco's synthetic-control fit, donors are restricted to the same state to share state-level shocks (such as the March–September 2022 national fuel-excise cut, which we have verified to be a symmetric common shock).

Outcome. Monthly mean unleaded retail price (regular ULP / U91 / E10) at the treated market or synthetic-counterfactual market, in cents per liter.

Identifying assumption. The synthetic version of each Costco's local market — constructed as a weighted combination of donor postcodes that closely matches the treated market's pre-treatment price trajectory — would have continued to track the treated market's outcome in the absence of Costco entry. We assess this assumption visually via pre-treatment fit plots and statistically via placebo synthetic controls run on every donor postcode in turn; the treatment effect is credible if its post-treatment magnitude exceeds the placebo distribution.